FixoraAI

Enterprise Multimodal Brand Governance & Content Publishing Platform

Hackathon Stage: Round 3 Final Submission (Top 50)	**Team Leader:** Pranamya Kalambe
Live Web Application: https://fixoraai.onrender.com	**Production API Docs:** https://fixoraai-backend.onrender.com/docs
Source Repository: https://github.com/Pranamya1/FixoraAI	**Evaluator Access:** 1-Click Judge Access (No sign-up needed)

Executive Overview

Modern marketing and engineering organizations face a critical governance dilemma: generative AI enables teams to produce content 100× faster, but leaves enterprise brands vulnerable to severe brand dilution, non-compliant color palettes, visual design errors, and legal claims. **FixoraAI** solves this by embedding an intelligent, multimodal governance layer directly between creative generation and social publishing.

Rather than functioning as a surface-level LLM wrapper, FixoraAI extracts and indexes company brand guidelines into vector embeddings, performs multimodal visual inspection using **Google Gemini Vision** on creative posters, routes approval requests to executives across four channels (WhatsApp, Gmail, Slack, Teams), and executes 1-click live publishing via real OAuth.

1. Delivered Round 3 Improvements (Committed at Round 2)

As committed in the Round 2 milestone submission, all three designated major engineering improvements have been delivered, tested, and deployed to production:

Delivered Improvement	Technical Implementation & User Value
Improvement 1: **Direct Multi-Platform Social Publishing**	• **Real LinkedIn OAuth 2.0:** Engineered live authorization code flow with secure token persistence in Supabase, enabling 1-click publishing directly to live LinkedIn feeds. • **Twitter/X & Slack Integration:** Twitter/X pre-formatted Web Intent dispatch and Slack channel webhook broadcasts. • **Third-Party API Constraints (Transparency):** LinkedIn is 100% live direct publishing. Twitter/X programmatic API write access requires a paid $100/mo enterprise tier, so FixoraAI utilizes 1-click Web Intent dispatch instead. Meta (Instagram & Facebook) requires multi-week Business Organization Verification and manual App Review, so FixoraAI provides verified brand handle linking and staged copy distribution.
Improvement 2: **Multimodal Creative Visual Auditing**	• **Gemini Vision Inspection:** Implemented deep visual inspection of generated posters and uploaded images to verify brand color hex tokens (e.g., #009DFF primary), typographical legibility, logo placement, and layout balance. • **Dual Scorecards:** Provides creators with instant visual compliance ratings alongside textual guideline checks.

Improvement 3: Executive Review & Radio Dispatch Hub	• **Single-Platform Radio Selector:** Redesigned dispatch modal to single-channel radio selection, preventing multi-channel notification spam. • **Cross-Platform Gateways:** Instant WhatsApp wa.me mobile chat links, executive responsive HTML emails, interactive Slack Block Kit cards, and Microsoft Teams Connector cards with custom webhook overrides.

2. Complete Production Feature Matrix

Category	Feature Name	Capabilities & Architectural Details	Status
Knowledge Ingestion	Brand Guideline Vector Parser	Extracts text and layout from enterprise PDFs/Markdown; chunks and generates 768-dim embeddings in Supabase pgvector.	**Production**
Compliance Engine	RAG Semantic Auditor	Audits drafts against indexed company policies with rule citations, severity classifications (Critical/Warning/Notice), and 1-click fixes.	**Production**
Multimodal AI	Gemini Vision Visual Inspector	Visual inspection of graphic layouts, contrast ratios, and color palette tokens (#009DFF) powered by Gemini Vision.	**Production**
Creative Studio	Multi-Deliverable Synthesis	Generates channel-calibrated copy, 1080×1080 branded social posters (.PNG via Pillow), and 4-slide PDF carousels (ReportLab).	**Production**
Review & Sign-Off	Executive Approval Dispatch	Single-platform radio dispatch across WhatsApp, Gmail HTML cards, Slack Block Kit, and Microsoft Teams cards.	**Production**
Social Publishing	Direct Channel Integration	Live OAuth 2.0 direct feed posting to LinkedIn; verified Twitter/X web intent; channel handle linking in settings.	**Production**
Authentication	1-Click Judge Access	Instant evaluator walkthrough button bypassing onboarding and credential friction while supporting full Supabase Auth.	**Production**

3. Technical Architecture & Implementation Summary

FixoraAI is engineered as a decoupled, high-throughput full-stack system built for low-latency governance verification:

- **Frontend Client:** React 19, TypeScript, and Vite styled with modern Tailwind CSS v4. State persistence is managed through resilient local storage synchronization and reactive event listeners.
- **Backend API:** Python 3.10+ with FastAPI utilizing asynchronous request handling, Pydantic data contract validation, and connection pooling for external services.
- **AI & Multimodal Core:** Google Gemini API (gemini-3-flash-preview for contextual reasoning and structured audit outputs; Gemini Vision for graphic and typography validation; models/embedding-001 for semantic retrieval).
- **Database & Vector Store:** Supabase PostgreSQL enhanced with the pgvector extension for sub-50ms cosine similarity searches against chunked brand guidelines.
- **Asset Synthesis Pipelines:** Headless Python graphics pipelines utilizing Pillow (PIL) for crisp 1080×1080 brand poster composition and ReportLab for structured, multi-page LinkedIn carousel PDFs.
- **Enterprise Integrations:** Native OAuth 2.0 protocol for LinkedIn live publishing, Webhook dispatchers for Slack/Teams, and dynamic deep-linking for WhatsApp and mailto clients.

4. Reliability & Independent Evaluator Testing

To satisfy the Round 3 requirement that *'a judge will open your link and use it on their own with nobody from your team explaining anything,'* FixoraAI incorporates comprehensive automated defensive engineering:

- **Zero-Setup Graceful Fallback:** If a judge begins generating content without uploading custom guidelines, FixoraAI automatically falls back to an enterprise default brand kit, preventing null-pointer errors or empty state blockers.
- **SPA Cloud Routing Protection:** Implemented a production _redirects rewrite policy (/* /index.html 200) to ensure deep links, settings tabs (/?tab=settings), and OAuth redirect callbacks never produce 404 Not Found errors on Render.
- **State Resilience on Browser Refresh:** Active review sessions, generated drafts, and user tokens persist across hard browser reloads through atomic storage synchronization.
- **Defensive Input Validation:** All API routes enforce strict schema boundaries via Pydantic; invalid phone formats, malformed webhook URLs, and empty topics display clean inline warnings rather than throwing unhandled 500 exceptions.
- **Build & Type Integrity:** 100% clean TypeScript compilation check (npm run build, zero errors across 2,100+ transformed modules) and clean Python backend import verification.

5. Product Roadmap & Future Commercial Development

Phase / Milestone	Key Deliverables & Business Impact
Q1 2027 **Enterprise SSO & RBAC**	Implement SAML 2.0 / Okta enterprise identity with hierarchical role-based permissions (Content Creator, Compliance Officer, Chief Brand Officer).
Q2 2027 **Video Brand Governance**	Extend multimodal inspection to short-form video (TikTok, Instagram Reels, YouTube Shorts) utilizing frame-by-frame audio and visual compliance scanning.
Q3 2027 **Automated Legal Localization**	Incorporate jurisdiction-aware legal compliance checks (FTC endorsement guidelines, GDPR cookie/privacy declarations, and ASCI advertising regulations).
Q4 2027 **Agency Multi-Brand Suite**	Develop a multi-tenant portal enabling marketing and PR agencies to govern and publish across hundreds of distinct client brand guidelines from a unified cockpit.

Submitted by: Pranamya Kalambe (Team Leader) | **Product:** FixoraAI | **Round:** 3 (Final)

Production URL: https://fixoraai.onrender.com | **Code Repository:** https://github.com/Pranamya1/FixoraAI